

Reviewer Pack — Minimal Order of Grothendieck-Serre Duality and Resolution P...

Entry c4012dfb56c4 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Grothendieck-Serre Duality and Resolution Proof Width

Entry ID: c4012dfb56c4 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-08 00:36:01 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Grothendieck-Serre Duality) **Field B** (complexity object): Boolean Satisfiability (Resolution Proof Complexity)

Statement:

For every Boolean formula φ with n variables, the minimal order of the Grothendieck-Serre dual object $G(\varphi)$ in the associated abelian variety is linearly correlated with its resolution proof width $w(\varphi)$, such that $\min_order(G(\varphi)) = \Theta(w(\varphi))$.

Rationale (proposer's reasoning):

Grothendieck-Serre duality provides a bridge between algebraic geometry and arithmetic complexity. It may reveal hidden structures in the algebraic properties of Boolean formulas, leading to insights into their computational hardness.

Taxonomy category: Algebraic Geometry - Resolution Proof Complexity Correlation (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 03eb8509723986ae

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a Boolean formula φ with n variables, if $\min_order(G(\varphi))$ is linearly correlated with $w(\varphi)$ such that $|\min_order(G(\varphi)) - \Theta(w(\varphi))| \leq 3$ for all seeds and the correlation coefficient is ≥ 0.9 .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	SAFE
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Grothendieck-Serre Duality" AND "resolution proof complexity" - "algebraic geometry" AND "Boolean satisfiability problem" - "minimal order" AND "abelian variety" AND resolution

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_formula(n):
        if n == 1:
            return 'A'
        else:
            left = generate_boolean_formula(n // 2)
            right = generate_boolean_formula(n - n // 2)
            return f'({left} OR {right})'

    def resolution_proof_width(phi):
        stack = []
        for char in phi:
            if char == '(':
                stack.append(char)
            elif char == ')':
                while stack[-1] != '(':
                    stack.pop()
                stack.pop()
            if len(stack) > 0 and stack[-1] == 'OR':
                stack.pop() # Remove the OR operator
        return len(stack)

    def frobenius_endomorphism_order(G):
        n = len(G)
        I = [[int(i == j) for j in range(n)] for i in range(n)]
        F = G.copy()
        order = 1
        while True:
```

```

        F = matrix_multiply(F, F)
        if matrix_equal(F, I):
            return order
        order += 1

def matrix_multiply(A, B):
    n = len(A)
    C = [[0 for _ in range(n)] for _ in range(n)]
    for i in range(n):
        for j in range(n):
            for k in range(n):
                C[i][j] += A[i][k] * B[k][j]
    return C

def matrix_equal(A, B):
    n = len(A)
    for i in range(n):
        for j in range(n):
            if A[i][j] != B[i][j]:
                return False
    return True

n = random.randint(5, 40)
phi = generate_boolean_formula(n)
w_phi = resolution_proof_width(phi)

G = [[random.choice([0, 1]) for _ in range(n)] for _ in range(n)]
min_order_G = frobenius_endomorphism_order(G)

return {
    "metric_name": "min_order(G( $\phi$ ))",
    "metric_value": min_order_G,
    "instances_tested": 1,
    "n_max": n,
    "conjecture_holds": False,
    "counterexample": "mapping_undefined"
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:

```

```

first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means it did not meet the pre-registered support condition for a clear SUPPORTED verdict. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14743
2	preregistration	ollama_remote	glm4:latest	0	0	9501
3	novelty	ollama_remote	glm4:latest	0	0	12529
4	novelty	ollama_remote	glm4:latest	0	0	9351
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	25636
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11324
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7993
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9569
9	judge	ollama_remote	glm4:latest	0	0	16019

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 116667 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/c4012dfb56c4.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/c4012dfb56c4.jsonl` for verification of provenance

4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/c4012dfb56c4.tar.gz (if generated)